

SPRINT DOCUMENTATION

Sprint 758 — Production Release Drift Detection

Objective

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Document Control

Purpose	Enterprise platform overview and buyer evaluation reference
Audience	Business leaders, platform administrators, evaluators, partners, and technical stakeholders
Scope	Capabilities, architecture, security, deployment, adoption, outcomes, and terminology
Source	ServicePro repository documentation; technical meaning preserved

Sprint 758 detects differences between the trusted release admitted by Sprint 757 and the release that is actually running in an environment.

The control is fail-closed. A deployment is considered aligned only when all required evidence fields exist and match.

Required evidence

The detector compares:

- `releaseId`
- `version`
- `sourceCommit`
- `artifactDigest`
- `environment`

Trusted evidence is loaded in this order:

- 1 `release-evidence/trusted-release-entry.json`
- 2 `release-evidence/trusted-release-registry.json`

The paths may be overridden with:

- `TRUSTED_RELEASE_ENTRY_PATH`
- `TRUSTED_RELEASE_REGISTRY_PATH`
- `RELEASE_EVIDENCE_DIR`
- `RELEASE_DRIFT_REPORT_PATH`

Actual runtime evidence may be supplied by `RUNTIME_RELEASE_EVIDENCE_PATH`, or with environment variables:

- `RELEASE_ID` or `SERVICEPRO_RELEASE_ID`
- `APP_VERSION` or `SERVICEPRO_VERSION`
- `GIT_COMMIT_SHA`, `RENDER_GIT_COMMIT`, `COMMIT_SHA`, or `SOURCE_COMMIT`
- `RELEASE_ARTIFACT_DIGEST`, `ARTIFACT_DIGEST`, or `RELEASE_BUNDLE_DIGEST`
- `DEPLOYMENT_ENVIRONMENT`, `RENDER_SERVICE_NAME`, or `NODE_ENV`

Status values

- aligned: all required fields exist and match.
- drifted: expected and actual values exist but one or more values differ.
- unknown: trusted evidence exists, but runtime evidence is incomplete.
- evidence_missing: trusted evidence is absent or incomplete.

Every state except aligned returns a nonzero process exit code.

Commands

```
npm run test:sprint758
npm run release:drift-check
```

The drift report is written to:

```
release-evidence/production-release-drift-report.json
```

The report contains fingerprints and comparison results. It does not write raw secrets, credentials, signing keys, or tokens.

Production example

```
$env:RELEASE_ID = 'release-2026-07-18'
$env:APP_VERSION = '8.0.0-alpha.1'
$env:GIT_COMMIT_SHA = '<deployed commit SHA>'
$env:RELEASE_ARTIFACT_DIGEST = '<trusted artifact digest>'
$env:DEPLOYMENT_ENVIRONMENT = 'production'

npm run release:drift-check
```

Rollback

Sprint 758 adds standalone scripts, tests, documentation, and package commands. To remove it before commit:

```
git restore package.json
Remove-Item scripts\check-production-release-drift.js -Force
Remove-Item scripts\lib\release-drift-detector.js -Force
Remove-Item tests\sprint758-production-release-drift-detection.test.js -Force
Remove-Item docs\sprint758-production-release-drift-detection.md -Force
```

After a commit, use `git revert <sprint-758-commit-sha>` rather than rewriting branch history.